

Motivation

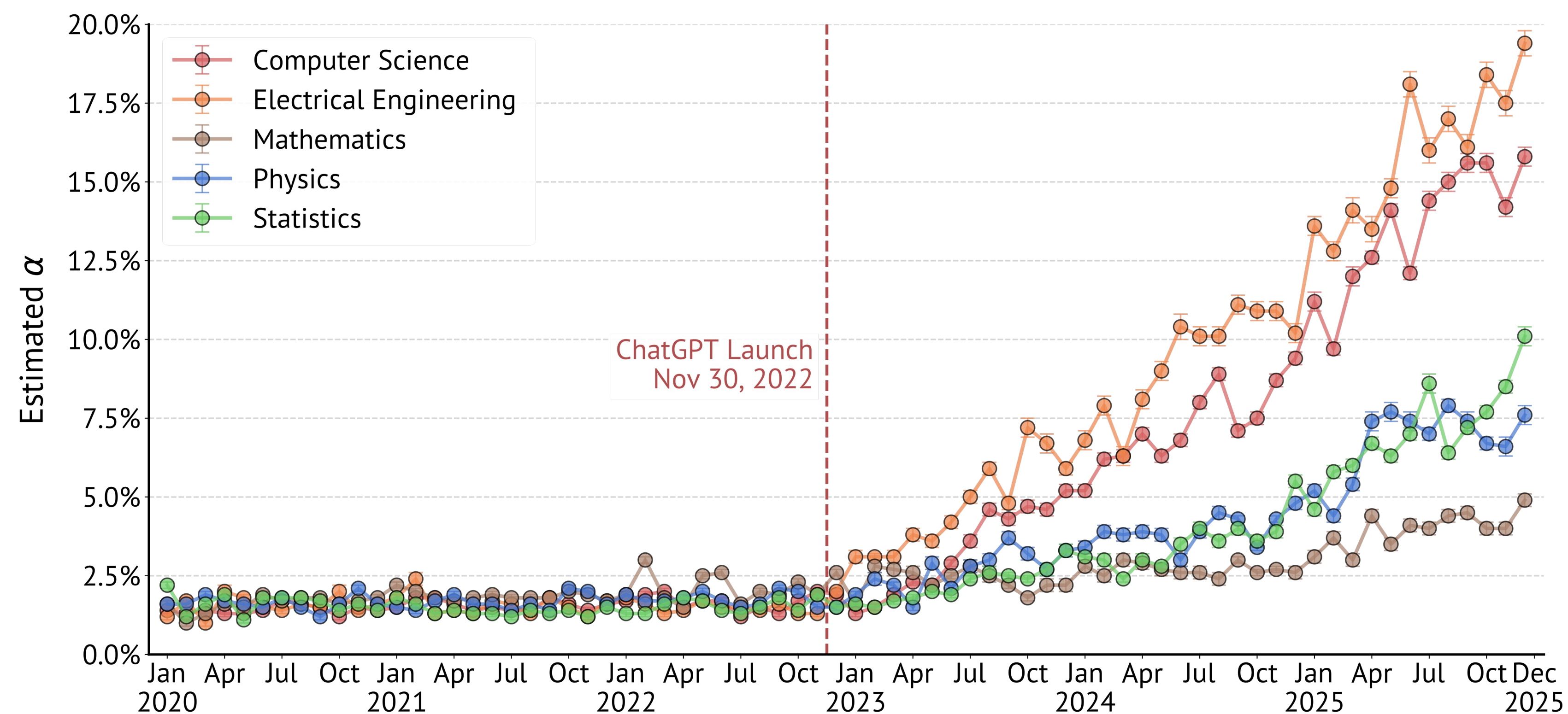
- Use ML models to quantify the proportion of LLM-modified text in research papers published after ChatGPT's release.
- Extend the scope of prior research on LLM use from abstracts and peer reviews to full text of articles.

Data

- Training: human-written ArXiv papers in 2020 and 2021 (Computer Science, Electrical Engineering, Mathematics, Physics, Statistics) and AI-generated versions of these papers.
- Validation: pre-mixed human and AI-generated sentences from ArXiv posted in 2022.
- Inference: ArXiv papers from 2020 through 2025.

AI Paper Generation Prompt

1. Compress
Reverse-engineer the source paper into a structured bullet-point outline, preserving all technical details, numerical results, and methodology. Flag formulas, tables, and citations as [FORMULA], [TABLE], and [CITATION].
2. Expand
Rewrite each outline section into formal academic prose, section by section, targeting the original word count. Do not invent equations, tables, or citations—reproduce placeholders verbatim.
3. Coherence Edit
Smooth transitions, unify terminology, and remove repetition across the assembled paper. Preserve all technical content exactly—add no new claims or meaning.
4. Title and Abstract Generation
Generate an 8–20-word academic title specific to the paper's main contribution. Write a 150–250-word self-contained abstract covering motivation, methods, results, and conclusions.
5. Judge Output
Score the paper on four dimensions (1–5): coherence, fluency, hallucination risk, and academic quality, plus an overall score and a list of flagged issues.



Methodology

1. Generate training data to create word frequency distributions for human and AI text.
2. Create mixtures of human and AI sentences at 11 different α levels to test and validate the model.
3. Use the model to predict the proportion of LLM-modified text (α) in unseen papers after 2022.

Main Takeaways

- LLM use in scientific papers has steadily increased since 2022.
- Analysis of the full-text resulted in a lower estimate of α compared to analysis of abstracts.
- Computer Science and Electrical Engineering showed the highest LLM use.

Future Work

- Use n-grams instead of individual words to compare human and AI-generated papers.
- Estimate α using Deep Learning models.

Reference

W. Liang *et al.*, “Quantifying large language model usage in scientific papers,” *Nature Human Behaviour*, Dec. 2025.